01 · LESSON

Reactions in plants

9B Plant growth · Lesson 1 of 7

Date _____ Class _____

Today I can

- ☐ I can write the word and symbol equations for photosynthesis and aerobic respiration, and say where and when each happens in a plant.
- ☐ I can explain how limiting factors slow photosynthesis and why roots need oxygen.

What happens today

1. Do Now: on a farm
2. Learn: Photosynthesis · Light, chlorophyll and limiting factors · Aerobic respiration · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 21 to 23 · Workbook pages 15 to 16

What I learned today (one or two sentences)

02 · KEY VOCABULARY

Key Vocabulary

Term	Definition
photosynthesis	the process plants use to make glucose from carbon dioxide and water, using light
aerobic respiration	the reaction in every living cell that releases energy from glucose using oxygen
chlorophyll	the green substance that captures energy from light
chloroplast	the part of a plant cell where photosynthesis happens
glucose	the sugar made in photosynthesis
reactant	a substance used up in a chemical reaction
product	a substance made in a chemical reaction
limiting factor	a variable that slows down the rate of photosynthesis
phloem	tubes made of living cells that carry sugar solution around a plant
yield	the amount of useful product a farmer gets from a crop

Quick Review

on a farm

Farmers want as much useful product (yield) from their crops as possible.

1 What process do wheat plants use to make their own food?

2 Name the gas that plants take in for this process.

3 Suggest the effect of a very cloudy year on the wheat harvest.

Core 1 of 3

Everyone does these. Answer on the card or under it.

1 Complete the word equations. Use the word bank.

Photosynthesis: _____ + water makes _____ + oxygen

Aerobic respiration: glucose + _____ makes carbon dioxide + _____ (+ energy released)

2 Tick one box in each row to show whether the statement is about photosynthesis or aerobic respiration.

Statement	Photosynthesis	Respiration
happens only when there is light		
happens all the time, day and night		
takes in carbon dioxide		
releases energy from glucose		
happens in chloroplasts		
happens in every living cell		

Core 2 of 3

Everyone does these. Answer on the card or under it.

3 Match each key word to its meaning. Draw a line.

chlorophyll	tubes of living cells that carry sugar solution around the plant
chloroplast	the sugar made in photosynthesis
limiting factor	the green substance that captures energy from light
phloem	a variable that slows down the rate of photosynthesis
glucose	the part of a plant cell where photosynthesis happens

4 Van Helmont grew a willow tree in a pot of dry soil for five years. He added only rainwater. The table shows his results.

	At the start	After five years
mass of dry soil (kg)	90.72	90.66
mass of tree (kg)	2.27	76.74

a) Calculate the change in the mass of the soil.

Core 3 of 3

Everyone does these. Answer on the card or under it.

4 *continued*

b) Calculate the change in the mass of the tree.

c) Aristotle thought that roots sucked up soil for the plant to eat. Explain how these results show that Aristotle was wrong.

Challenge

When Core is finished. Full sentences.

C1 Draw a flow chart to show how energy from the Sun becomes stored in a plant. Use the words light, chlorophyll, chloroplast and glucose.

Draw here.

C2 After his experiment, van Helmont concluded that all the extra mass of the tree came from the water he added. Evaluate this conclusion. Use the word equation for photosynthesis in your answer.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Write the word equation for photosynthesis.

2 Where in a plant cell does photosynthesis happen?

3 Why do plants in a field only photosynthesise during the day?

Today I can: Goal 1 ☐ Goal 2 ☐